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Review

Recent Advances in Remote Sensing and Artificial Intelligence for River Water Quality Forecasting: A Review

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Abstract: Rapid population growth and climate change have created challenges for managing water quality. Protecting water sources and devising practical solutions are essential for restoring impaired inland rivers. Traditional water quality monitoring and forecasting methods rely on labor-intensive sampling and analysis, which are often costly. In recent years, real-time monitoring, remote sensing, and machine learning have significantly improved the accuracy of water quality forecasting. This paper categorizes machine learning approaches into traditional, deep learning, and hybrid models, evaluating their performance in forecasting water quality parameters. In recent years, the long short-term memory (LSTMs), gated recurrent units (GRUs) and LSTM- and GRU-based hybrid models have been widely used in forecasting inland river water quality. Combining remote sensing with a real-time water quality monitoring network has enhanced data collection efficiency by capturing spatial variability within the river network, complementing the high temporal resolution of in situ measurements, and improving the overall robustness of predictive deep learning models. Additionally, leveraging weather prediction models can further enhance the accuracy of water quality forecasting and better decision-making for water resource management.

Keywords: machine learning; water quality management; river water quality forecasting

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Academic Editor: Guobin Fu

Received: 9 April 2025
Revised: 3 May 2025
Accepted: 8 May 2025
Published: 10 May 2025

Citation: Pan, D.; Deng, Y.; Yang, S.X.; Gharabaghi, B. Recent Advances in Remote Sensing and Artificial Intelligence for River Water Quality Forecasting: A Review. *Environments* **2025**, *12*, 158. https://doi.org/10.3390/environments12050158

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1. Introduction

Water quality is a critical indicator of ecosystem health, human well-being, and socioeconomic development. Inland rivers, as vital freshwater sources, play a fundamental role in supporting ecosystems and providing water for agriculture, industry, and human consumption. However, concerns over the deterioration of river water quality have grown significantly in recent decades, driven by both natural and human factors [1–13].

1.1. Problem Statement

Major contributors to inland river water pollution include industrial discharge, agricultural runoff, urbanization, and climate change [13–17]. Untreated or poorly treated effluents from manufacturing, mining, and chemical industries introduce hazardous pollutants into rivers [14]. Through the widespread use of fertilizers and pesticides, agricultural activities lead to nutrient pollution and trigger eutrophication and harmful algal blooms [18–21]. Rapid urban growth increases domestic wastewater discharge and stormwater runoff [22]. Meanwhile, variations in rainfall patterns, rising temperatures, and extreme weather events further exacerbate water quality issues by altering river flow regimes and pollutant transport mechanisms [13,23–25].

These factors have degraded river ecosystems, reduced aquatic biodiversity, and created dead zones that severely impact aquatic life [26,27]. Consequently, there is an urgent need to develop accurate and efficient inland river water quality forecasting models to support proactive water resource management.

Between 2000 and 2010, traditional machine learning models were widely used in river water quality forecasting. However, traditional water quality assessment methods relating to physical sampling and analysis are often time-consuming, costly, and geographically limited. After that, big data techniques and neural networks also started gaining traction in this domain. By the 2020s, deep learning models emerged as the most popular choice, owing to their superior performance in handling spatial and temporal river datasets. More recently, hybrid models have surpassed standalone deep learning approaches, offering greater accuracy and efficiency. These advanced hybrid models are now becoming the dominant choice in river water quality forecasting. This progression, synthesized from a critical review of the recent literature [28,29], highlights the evolving methodological landscape in river water quality forecasting.

In recent years, machine learning techniques have emerged as effective tools for forecasting river water quality because they can process large datasets and non-linear relationships to provide timely predictions [30,31]. While significant progress has been made, the current body of research lacks a comprehensive synthesis of the methods applied, as well as their respective strengths, limitations, and areas for improvement.

1.2. Contribution

This review focuses on inland rivers, including a wide range of river types from major river basins to smaller rivers, streams, and tributaries. It narratively categorizes machine learning approaches for inland river water quality forecasting, including traditional, deep learning, and hybrid models. It analyzes the characteristics, limitations, and challenges of recent methods, guiding the selection of model inputs tailored to river system features. Furthermore, it proposes future research directions such as hybrid model integration, addressing data sparsity through real-time forecasting, incorporating remote sensing for near-real-time monitoring, and enhancing model interpretability for decision-making support. By synthesizing current research, this review fosters the development of robust, scalable tools for sustainable water quality management, and lays the groundwork for future innovations in the field.

2. Materials and Methods

In inland river water quality forecasting, key elements include essential water quality indicators and performance metrics. The water quality indicators are categorized into physical, chemical, and biological parameters. Additionally, this paper highlights commonly used performance indicators to evaluate forecasting models effectively.

To ensure the review captures the latest advancements, the papers collected were mainly published between 2000 and 2025, reflecting the most recent research in the field. However, foundational and highly cited studies published before 2000 and those from 2000 to 2020 were also incorporated to provide historical context and deepen insights into the evolution of water quality forecasting methodologies. With the rise of machine learning, machine learning models have been commonly applied across many fields, including river water quality prediction.

A comprehensive literature search was conducted using Google Scholar to identify relevant studies published until 2025. Figure 1 shows the process of the review. The search strategy incorporated a combination of keywords, including “real-time monitoring water quality”, “inland river water quality forecasting”, “water management”, “inland river

water quality deep learning", "water quality hybrid machine learning", "water quality machine learning", "water quality urbanization agriculture", and "water quality remote sensing" .

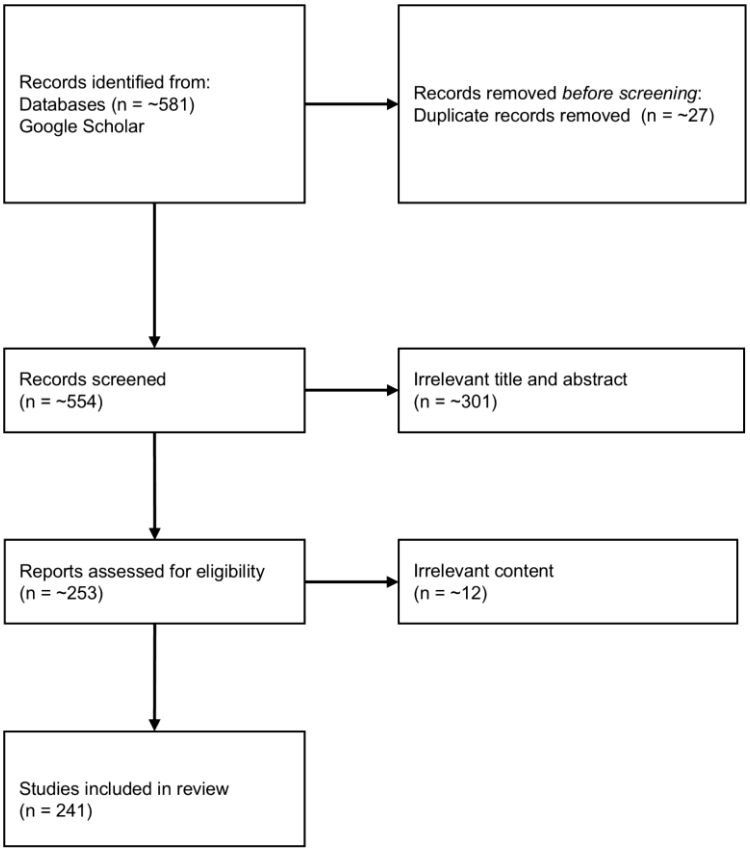


Figure 1. The process of the review. This diagram illustrates the conceptual logic of literature selection for this narrative review. The numerical values are approximate and not derived from a formal systematic review process.

In addition to studies directly applying machine learning techniques for inland river water quality forecasting, additional references were incorporated during the writing process to provide essential background information. These included studies related to water quality indicators, pollution sources, hydrological processes, and remote sensing technologies. As a result, the final synthesis reflects foundational knowledge necessary for a comprehensive understanding of the topic.

2.1. Hydrology Background

River water quality detection is important for water management and protects water resources. Water quality forecasting offers an early warning for polluted water bodies. For example, the dynamic forecasting model helps determine whether the water body has a high risk of pollution. Water management has become an urgent issue due to the increasing pressure from human activities, such as urbanization and globalization, and the impacts of climate change on water resources [32,33]. Water is crucial for agriculture and energy production, while urban and industrial sectors continue to drive growing demand, further intensifying the need for effective water resource management [33].

Figure 2 presents a conceptual diagram illustrating the key factors influencing inland river water quality. As depicted, three primary drivers—weather, urbanization, and agriculture—play significant roles in shaping water quality dynamics [34–37]. Section A outlines the water quality impacts driven by precipitation. During periods of low precipi-

tation and drought, river discharge decreases. Under such conditions, pollutants such as nutrients and heavy metals tend to accumulate due to limited dilution and slower flow. This accumulation can promote eutrophication and lead to increased concentrations of dissolved and suspended solids in the water column. Human activities are a major contributor to these changes, influencing nutrient loading, pollution levels, and hydrological conditions. Sections B and C highlight the impacts of urbanization and agricultural activities, respectively. Urbanization increases domestic wastewater discharge and industrial effluents, while agriculture contributes to nutrient enrichment through fertilizer runoff and animal waste. Although the climatic influences are not explicitly illustrated in Figure 2, they are further discussed in the following paragraph, with particular attention to the impacts of ice cover and road salt on river water quality.

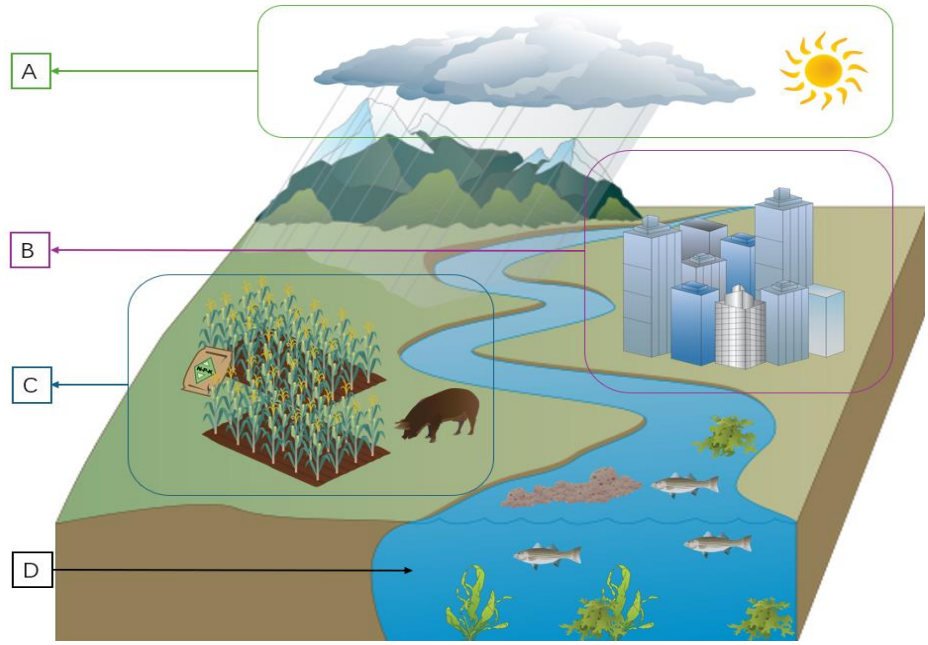


Figure 2. Conceptual diagram of river water quality influences. The labels refer to the following: (**A**) Increased river discharge, resulting from precipitation, leads to reduced turbidity and residence time, and consequently increases dissolved oxygen levels. (**B**) Urbanization increases pollution inputs, elevating nutrient levels, suspended solids, and organic contaminants. (**C**) Agricultural runoff delivers fertilizers and pesticides into rivers, leading to nutrient enrichment, turbidity increase, and dissolved oxygen decrease. (**D**) Eutrophication promotes algal blooms, which consume dissolved oxygen, causing hypoxia and the death of aquatic life. Figure made using icons from Integration and Application Network (ian.umces.edu/media-library; accessed on 9 May 2025).

In northern areas, seasonal ice formation significantly impacts river water quality. During winter, ice cover restricts gas exchange between water and air, leading to oxygen depletion. Moreover, ice formation alters turbidity and algal dynamics, which are key indicators of water quality [38,39]. Notably, the widespread use of road salt to accelerate ice melt leads to elevated chloride concentrations, which can disrupt aquatic ecosystems [40,41]. These seasonal variations underscore the complex interplay between climatic factors and river water quality, warranting further investigation into predictive modeling and management strategies.

Human activities contribute to water pollution, significantly impacting water quality. Pollution is generally categorized into two types—diffuse pollution and point source pollution [42]. Diffuse pollution is primarily caused by surface runoff and interflow, while wastewater emissions represent point source pollution. Unlike point source pollution, diffuse pollution arises over a broad area, making it more challenging to monitor and

control. Multiple factors influence river water quality forecasting, including the interaction between groundwater and surface water. For instance, rainfall can transport fertilizers into the soil, where they gradually infiltrate into groundwater [43]. When groundwater interacts with surface water, these nutrients are transferred into rivers, impacting water quality and potentially leading to eutrophication.

2.2. Water Quality Indicators

Water quality indicators are measurable parameters that provide critical information about water's physical, chemical, and biological characteristics. These indicators are used to assess, monitor, and predict the condition of river systems. Accurate water quality forecasting heavily relies on selecting and analyzing the most relevant indicators. Water quality indicators serve as fundamental parameters in river water quality forecasting, providing essential insights into aquatic ecosystems' physical, chemical, and biological conditions. Temperature, discharge, turbidity, total dissolved solid (TDS) and total suspended solids (TSS) are regarded as physical indicators. Dissolved oxygen (DO), biological oxygen demand (BOD), chemical oxygen demand (COD), pH, nutrient levels and heavy metals are chemical indicators. The biological indicators include algae and coliform bacteria amount [44]. Table 1 introduces detailed information for each indicator. While this table summarizes the most frequently reported water quality parameters, future investigations could extend the scope by including a broader range of dissolved organic compounds. Other dissolved organic components, such as artificial sweeteners and iodinated X-ray contrast media, antibiotics and pesticides, are also important water quality indicators. These compounds primarily originate from wastewater discharge and runoff, and their concentration can serve as useful markers for detecting pollution in inland river systems [45,46].

Table 1. Detailed information for water quality indicators.

Type	Indicators *	Core Function and Impact	Key Influencing Factors
Physical indicators	Temperature	Influences chemical, biological, and other physical indicators [23,47] (higher temperature causes lower DO [48])	Geographical factors and water depth
	Discharge	Affects pollutants (high discharge decreases concentrations of contaminants like nutrients and heavy metals [49,50]).	Rainfall and climate change
	Turbidity	Cloudiness of water, some suspended particles in water [19], [51–57]; reduces light penetration, limits primary production, and decreases DO.	Potential contamination and soil erosion.

Table 1. *Cont.*

Type	Indicators *	Core Function and Impact	Key Influencing Factors
Chemical indicators	TDS	Important drinking water quality indicators [58], indicating the exchange between groundwater and surface water [59]. Higher turbidity may cause higher TDS.	Geological composition, climate, and anthropogenic activities [60,61].
	TSS	The solid particle concentration suspended in water [55,62,63]; important for aquatic ecosystems' health and closely related to turbidity [57].	Climate, and anthropogenic activities
	DO	Important for the health and sustainability of aquatic ecosystems [64–67]. A lower DO level (less than 2 mg/L) shows hypoxic and dead zone [48]. Many water bodies have faced hypoxia [68–72].	Temperature and algae bloom
	BOD	The amount of DO aerobic microorganisms require to break down organic matter present [73–75], also an indicator of eutrophication.	Pollution release and organism consumption
	COD	The level of pollution in water bodies [76,77]; includes biodegradable and non-biodegradable compounds; a faster alternative to BOD for assessing overall pollution levels.	Industry waste and surface runoff
	pH	Aquatic life suffers from extreme pH values [55,78]. An extreme pH value threatens the life safety of most aquatic organisms.	Acid rain [79], industrial discharges and agricultural runoff
	Nutrient levels	Water nutrient levels (total nitrogen (TN), total phosphorus (TP) and ammonia) support the growth of plants and algae [19,20,80]. Excessive nutrient concentrations lead to eutrophication, harmful algal blooms, and oxygen depletion.	Fertilizer application, domestic sewage
	Heavy metals	Toxic (lead (Pb), mercury (Hg), cadmium (Cd), chromium (Cr) [54]) even when concentrations are low [81,82].	Industry waste
Bio indicators	Algae	Algal blooms disrupt the ecosystem balance. Some harmful algae produce toxins [83,84]. Algae amounts are determined by the amount of Chlorophyll-a (Chl-a) [85].	Nutrient levels [19,20,80,86], temperature and turbidity.
	Coliform bacteria	Live in warm-blooded animals' body, soil, and water [34,52,87–89]. A higher number of coliform bacteria causes higher BOD and lower DO.	Fecal matter, agriculture and pollution

* The parameters listed in this table reflect the most commonly monitored water quality indicators in river water quality forecasting studies. Future research may consider including additional dissolved organic compounds to further expand the assessment framework.

Water quality indicators exhibit both temporal and spatial variations and are influenced by other environmental factors. The table shows that temperature and turbidity affect DO, while discharge impacts turbidity. In water quality forecasting, correlation analysis is

commonly used to determine relationships between different indicators. Indicators with higher correlation coefficients are considered more influential in predicting the target water quality parameter [48], making them essential for improving forecasting accuracy.

These are the typical water quality indicators used in recent research. However, water quality indicators are inherently variable, as all components entering the water system contribute to changes in water quality. The following section provides an overview of recent advancements in river water quality forecasting.

2.3. River Water Quality Forecasting in Recent Years

Traditional monitoring methods rely on physical sampling and laboratory analysis, which are time-consuming, costly, and spatially limited, making it difficult to provide real-time assessments and early warnings for pollution events [30]. Furthermore, using machine learning, river water quality forecasting supports proactive decision-making in pollution control and disaster management. By identifying trends and predicting potential contamination events, machine learning models help policymakers implement targeted mitigation strategies, optimize wastewater treatment, and ensure drinking water supply safety [64]. Integrating hybrid machine learning models, combining deep learning and physics-based hydrological models, has further improved forecasting capabilities by capturing both temporal and spatial variations in river ecosystems [66].

Figure 3 shows that weather conditions, diffuse pollution, and point source pollution influence river water quality. Machine learning methods use remote sensing, discrete sampling, and real-time monitoring datasets to forecast river water quality. The predicted water quality data are then utilized in decision-support systems (DSS) to inform decision-making. Water management strategies, including policy development, are based on insights derived from DSS, enabling more effective and sustainable water resource management.

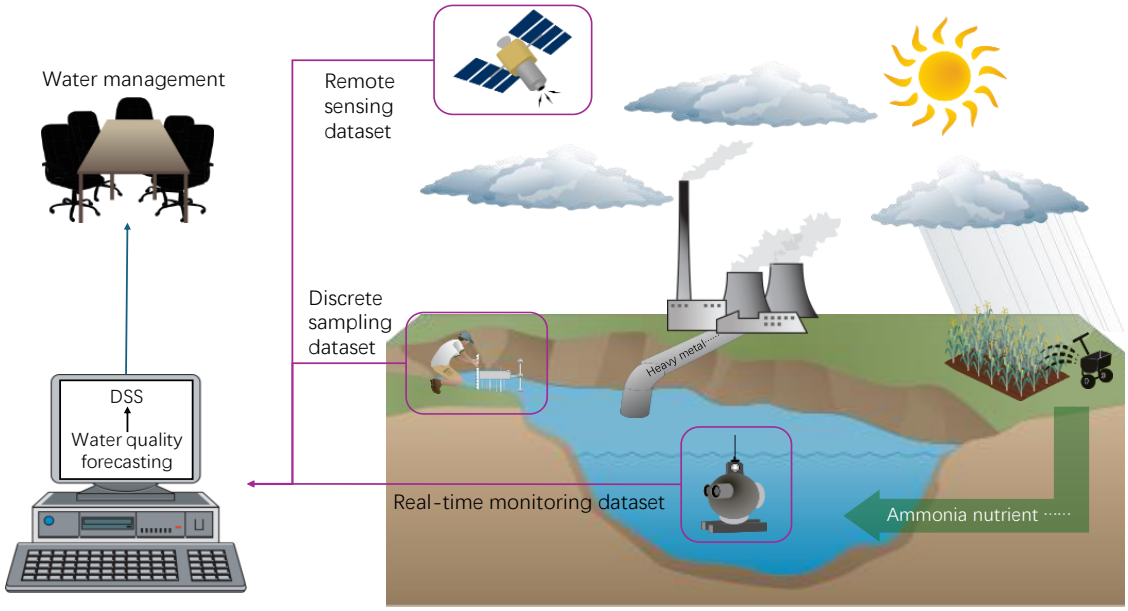


Figure 3. A schematic diagram of water quality monitoring. Figure made using icons from Integration and Application Network (ian.umces.edu/media-library; accessed on 9 May 2025).

Traditional methods face challenges such as the lack of real-time data, limited large-scale coverage, and low efficiency. Recent advancements incorporating real-time monitoring, remote sensing, and DSS help address these issues, enhancing the performance and effectiveness of water quality assessment and management.

2.3.1. Real-Time Monitoring for Water Quality

Real-time monitoring enables the collection of continuous, high-frequency water quality datasets [90], identifying water quality trends with greater accuracy. These datasets are particularly valuable for spatial models [91]. While real-time monitoring provides strong temporal resolution and operational efficiency [92], it may have some spatial limitations due to localized sensor deployment.

Some common indicators that apply real-time monitoring include TDS [93–95], temperature [93,96], pH [93,96], DO [89], and turbidity [96]. Monitoring spikes in these parameters can help detect various upland sources of industrial effluent discharges, agricultural non-point source pollution runoff, municipal wastewater discharges in combined sewer systems and stormwater runoff pollution to urban streams using well-established modelling tools [97]. These indicators provide an instant reflection of water quality changes, offering real-time insights into the health of aquatic ecosystems. Implementing real-time monitoring for these parameters enhances assessment and response to fluctuations in water quality, ensuring more effective water resource management and ecosystem protection.

2.3.2. Advances in Remote Sensing and Machine Learning for River Water Quality Prediction

River water quality prediction is essential for environmental management and public health. Integrating remote sensing and machine learning offers a promising alternative, enabling large-scale assessment of key water quality parameters.

Remote sensing has facilitated the collection of large-scale datasets, such as chlorophyll-a measurements spanning from 1970 to 2023 [98,99]. Traditional in situ sampling often requires more than a day to produce comprehensive contamination mapping. In contrast, the integration of satellite data with machine learning significantly reduced detection times to within a few hours [98]. As shown in Figure 4, dating back to the 1970s, the launch of Landsat-1 enabled the first satellite-based observations of inland and coastal waters [100]. Early studies demonstrated that satellite spectral measurements could be empirically correlated with water quality indicators such as turbidity and chlorophyll-a, establishing remote sensing as a viable supplement to in situ sampling. Throughout the 1980s, advancements in multispectral sensors, such as the Landsat Thematic Mapper, expanded spectral coverage and spatial resolution, enabling more accurate access to water quality parameters [101]. The 1990s saw the introduction of new satellites, including SPOT and MODIS [102], alongside hyperspectral imaging [103], significantly enhancing the ability to detect subtle water quality signals. Concurrently, retrieval models evolved from purely empirical regressions to semi-analytical approaches [104] that integrated optical physics, ultimately leading to fully analytical (mechanistic) radiative transfer models. By the early 2000s, dedicated satellite missions, including MERIS and MODIS, were operational, and improvements in inversion algorithms, supported by increased computational power, yielded more accurate and globally applicable water quality estimates. In the 2010s, machine learning methods began to be applied to water quality prediction [105,106], capturing complex nonlinear relationships between spectral reflectance and water quality parameters. These data-driven approaches outperformed traditional empirical and semi-analytical models in predictive accuracy.

More recently, deep learning techniques have emerged, leveraging high-dimensional data from hyperspectral imagery and temporal records to further enhance retrieval accuracy and enable near-real-time monitoring. Such advancements underscore the disruptive potential of merging spectral data with computational intelligence. This section examines three pivotal innovations that are reshaping the field while addressing unresolved technical bottlenecks.

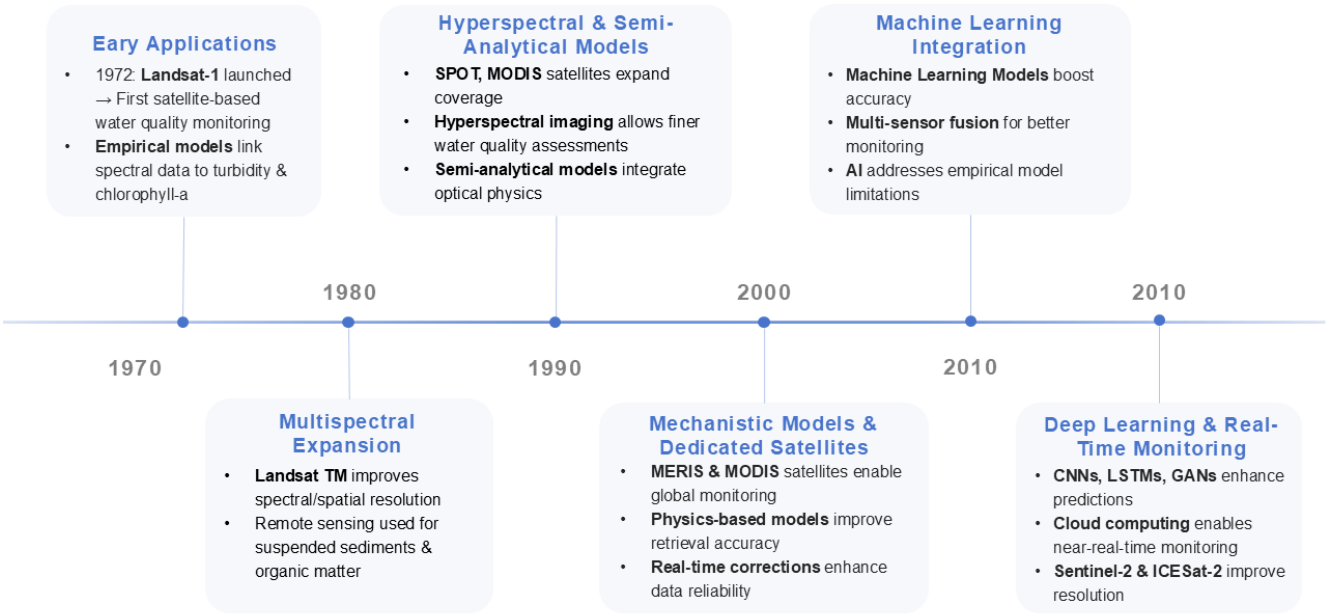


Figure 4. The timeline of remote sensing and machine learning for water quality monitoring.

Multi-sensor data fusion for water quality prediction: Multi-sensor data fusion made river water quality prediction more accurate. Chen et al. [107] demonstrated that UAV-based multispectral imaging combined with CatBoost regression performed better than traditional machine learning models in predicting turbidity, TN, and TP. Cheng et al. [108] further improved prediction accuracy using a stacked ensemble learning model (Stacked-RF), integrating gradient boosting, CatBoost, and Adaboost techniques. Despite these advancements, challenges remain in synchronizing multi-sensor data and addressing discrepancies in spatial and temporal resolutions. Future work should focus on optimizing sensor calibration and data harmonization techniques.

Innovations in spatiotemporal modeling: Spatiotemporal modeling is important in addressing the dynamic nature of river water quality. Zhu et al. [109] proposed a sub-regional inversion method using Gaofen-1 satellite data, significantly enhancing prediction accuracy in complex river networks. Zhao et al. [110] utilized Google Earth Engine (GEE) with a stacked generalization learning approach, achieving 91.67% accuracy in water quality classification for the Songhua River Basin. While these models enhance prediction reliability, their computational complexity and generalizability across diverse hydrological conditions remain challenges. The further exploration of hybrid physics-based deep learning models may address these limitations.

Few-shot learning for water quality prediction: It addresses data scarcity issues in river water quality prediction. Villota-González et al. [111] achieved R^2 scores of 0.72 for total suspended solids estimation using superlearner algorithms on limited Landsat-8 and Sentinel-2 datasets. Further, transfer learning approaches, such as those by Ueda et al. [112], demonstrated that for four peak water-level scenarios, the transfer learning model delivered 6 h forecasts with a Nash–Sutcliffe efficiency (NSE) of 0.86, matching the performance of traditional upstream-based methods (NSE = 0.84). Notably, prediction accuracy remained stable when training data for specific storm events were reduced from 12 to 3 periods, highlighting the framework's resilience in data-limited conditions. These approaches effectively circumvent data poverty but face trade-offs: model adaptation to new regions typically requires retraining 65–70% of network layers, eroding computational efficiency.

The selection of remote sensing platforms for river water quality monitoring must align with the spatial and temporal resolution requirements of target river reaches. For

instance, Landsat-8/9 (30 m spatial resolution, 16-day revisit) is well-suited for large rivers (>50 m width) to monitor chlorophyll-a (Chl-a) and turbidity trends, as demonstrated in nutrient monitoring studies for major basins [107,113]. Sentinel-2 (10–60 m resolution, 5-day revisit) offers a balance between spatial detail and temporal frequency, making it ideal for medium-sized rivers (20–50 m width) to track algal bloom dynamics and suspended sediment transport [107]. In contrast, MODIS (250–1000 m resolution, daily coverage) provides macroscale insights for estuaries and large river systems, despite its limited ability to resolve narrow channels [107,114]. Emerging technologies like UAV-based hyperspectral imaging (0.1–1 m resolution) enable the high-precision mapping of pollution sources in small tributaries or urban channels, such as pinpointing industrial discharge points or localized nutrient hotspots [115]. This multi-scale synergy ensures comprehensive monitoring, where coarse-resolution satellites capture basin-wide trends, while UAVs and high-resolution sensors address site-specific challenges [113].

The focus on remote sensing in this review stems from its transformative ability to overcome limitations inherent to traditional monitoring methods. Unlike labor-intensive in situ sampling, which lacks spatial coverage and temporal continuity, remote sensing enables large-scale, real-time assessments of optically active parameters (e.g., Chl-a, turbidity) across entire watersheds [114]. For example, Sentinel-2's frequent revisit cycle allows the near-real-time detection of harmful algal blooms (HABs), triggering timely interventions to mitigate public health risks [99,113]. Furthermore, remote sensing reduces operational costs by up to 60% compared to manual sampling, particularly in remote or hazardous regions [113]. While traditional methods remain critical for ground-truth validation, remote sensing provides unparalleled scalability for global water quality management. Future advancements in hyperspectral sensors (e.g., PRISMA, DESIS) and AI explainability frameworks will further bridge gaps in monitoring non-optically active parameters, solidifying remote sensing as a cornerstone of sustainable water resource management.

Although this review focuses on remote sensing methods due to their broad spatial coverage and scalability, it is also recognized that artificial intelligence (AI) models have proven effective when applied to high-frequency water column observations collected via sensor networks [116]. Combining those two methods will improve predictions across diverse spatial and temporal scales.

Integrating remote sensing and machine learning has significantly advanced river water quality prediction, providing unprecedented spatial and temporal coverage. Despite these advances, challenges remain in optimizing remote sensing methodologies, improving model generalization, and addressing discrepancies in multi-sensor data integration. Continued research is necessary to refine hybrid modeling approaches that incorporate physics-based constraints, enhance the interpretability of deep learning predictions, and develop more robust calibration techniques for non-optically active water quality parameters.

2.3.3. Decision-Support Systems for Water Management

DSS for water management is a system applied to water resources and datasets to make decisions [117]. These water resources include water quality datasets from predictive models and weather variations. DSS is crucial in decision-making, policy formulation, and in guiding research in related fields [118]. One example of DSS implementation is the SMART method [119]. The SMART method is successfully applied to issue warnings for the Bedog and Sembung Rivers, where water quality is deemed unsuitable for daily use. Another successful DSS application is the WATERMAN DSS, which integrates geographical information systems (GIS) with satellite imagery to effectively monitor the Strymon River [120].

AI DSS has become a prominent research topic in recent years. Unlike traditional DSS, AI DSS incorporates machine learning techniques, including deep learning methods, to generate datasets for decision-making. These AI-driven models enhance efficiency and reduce operational costs [121]. In water quality management, AI DSS utilizes AI models to predict water quality indices, enabling more accurate assessments. AI DSS and DSS are particularly effective in handling complex, variable, and extreme water bodies [122,123].

Despite these advancements, accurately forecasting water quality trends remains a complex challenge, as it involves nonlinear interactions between physical, chemical, and biological processes influenced by climatic variability, human activities, and hydrological dynamics. To address these challenges, machine learning models are powerful tools for water quality prediction, offering the ability to detect patterns in large datasets and provide highly accurate, data-driven forecasts.

3. Machine Learning Models Used in River Water Quality Forecasting

Machine learning models have been increasingly applied to forecast inland river water quality in recent years. As a vast and dynamic field, machine learning encompasses a wide variety of models.

As presented in Figure 5, the river water quality forecasting process is mainly divided into four steps. The methods of data preprocessing and model development have been continuously evolving in recent years, incorporating advanced techniques to improve accuracy and efficiency in water quality forecasting.

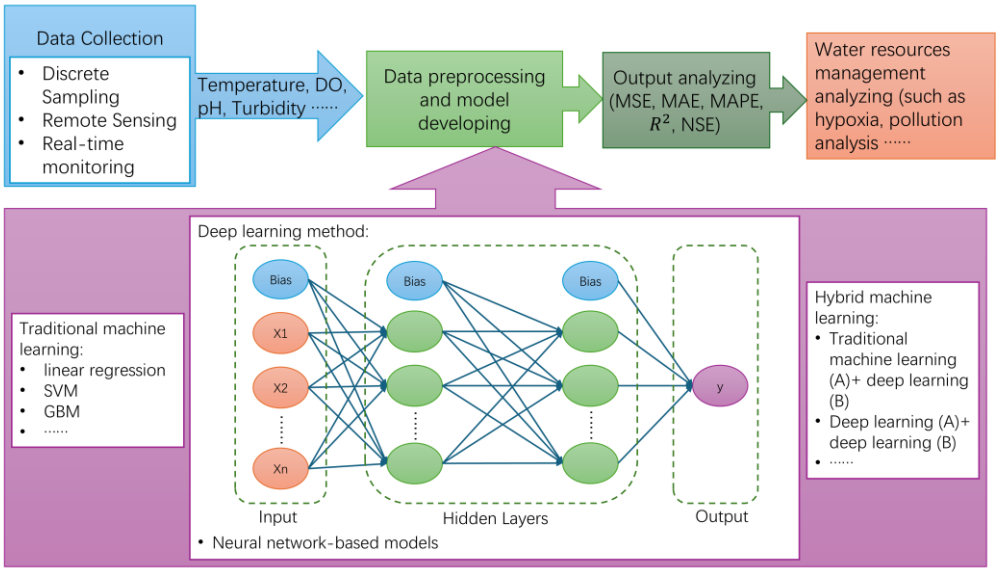


Figure 5. The flowchart of river water quality forecasting.

Besides the model development process, the output analysis is also important for water quality forecasting. In this step, performance indicators are metrics used to evaluate machine learning models' effectiveness, accuracy, and reliability. These indicators are crucial for assessing how well models predict parameters in river water quality prediction.

Mean square error (MSE) highlights model performance, especially in the presence of outliers. The root means squared error (RMSE) expresses errors in the same units as the target variable, enhancing interpretability. Mean absolute error (MAE) is more robust against outliers, making it a reliable accuracy measure. Mean absolute percentage error (MAPE) provides insights into relative performance. Lower values for MSE, RMSE, MAE, and MAPE indicate better model effectiveness, ensuring clarity and confidence in real-world applications.

The Coefficient of Determination (R^2) is close to 1, which indicates better performance. NSE is commonly used in hydrology to evaluate prediction accuracy relative to the mean observed value. NSE measures predictive skill relative to observed variability. R^2 measures the goodness of fit and explained variance.

Understanding the performance indicators helps to determine the machine learning performance. This section illustrates the machine learning models used nowadays for river water quality forecasting. Three primary categories of models stand out: traditional machine learning models, deep learning models, and hybrid machine learning models. This section highlights examples of machine learning applications in predicting river water quality.

3.1. Traditional Machine Learning Models

Traditional machine learning models were popular for river water quality prediction because they can process structured datasets, uncover patterns, and make predictions. Unlike deep learning models, traditional machine learning uses manual feature engineering and is generally more interpretable. It is particularly effective for datasets with limited complexity or size, making it useful for many water quality prediction tasks.

Multi linear regression models are the fundamental models used in river water quality forecasting [124]. As predicted, river water quality over a certain period of time is a regression problem. Support vector machines (SVMs) use hyperplanes to classify data points [125,126]. They can predict discrete water quality classes. The gradient boosting machines (GBMs) sequentially build an ensemble of weak learners, focusing on correcting errors from previous iterations. XGBoost, LightGBM and CatBoost are popular variants of GBMs [127–129].

Traditional machine learning models are relatively simple, and they can be explained. They are effective for small datasets. However, traditional machine learning methods are not as popular as complex machine learning methods as more machine learning models are being developed.

3.2. Deep Learning Models

Deep learning has revolutionized the field of data-driven modeling by effectively handling large, complex and unstructured datasets. Deep learning models in river water quality prediction excel at capturing non-linear relationships and temporal dependencies between multiple water quality indicators, making them particularly suitable for dynamic and complex river ecosystems.

They can handle complex datasets better than traditional machine learning models. They are used for high-dimensional datasets with different temporal and spatial information. In recent times, the most common deep learning models include artificial neural networks (ANNs), recurrent neural networks (RNNs), convolutional neural networks (CNNs), long short-term memory networks (LSTMs) and transformer models.

3.2.1. Artificial Neural Networks

ANNs are the foundational deep learning models. They are highly versatile and have been applied in water quality prediction due to their ability to capture non-linear relationships in data [130–135]. ANNs predict critical water quality indicators such as DO, BOD, pH, and turbidity [136–139]. ANNs analyze large and high-dimensional spatial and temporal water quality data to find pollution sources and predict their downstream impacts. This makes them suitable for modern water monitoring systems involving remote sensing [140].

ANNs are flexible and can be tailored to various prediction tasks, including regression and multi-output problems. They automatically learn relevant relationships from raw data,

reducing manual preprocessing demand. However, the accuracy of the ANN-predicted water quality parameters depends on the availability of a comprehensive and extensive monitoring program to provide a suitable training/testing dataset, which should include extreme events and significant effluent discharges and spills that would cause spikes in water quality parameters.

3.2.2. Recurrent Neural Networks

RNNs are designed to process sequential and time-series data. RNNs incorporate feedback loops, allowing them to maintain information about previous inputs (memory) in their computations. This makes RNNs particularly suitable for water quality prediction, where the temporal relationships between parameters are critical [141,142].

RNNs have a similar structure to ANNs. The input layer accepts sequential data such as time-series measurements of water quality parameters. The recurrent hidden layers are more complex than ANNs [143]. The hidden layers process inputs sequentially, using feedback loops to maintain a state that captures previous time step information. Each hidden state is computed as

$$h_t = f(W_{xh}x_t + W_{hh}h_{t-1} + b_h) \quad (1)$$

where h_t is the hidden state at the time step t, x_t is the input, and $f(\ )$ is an activation function. The output layer predicts the current or next time step for different water quality research. RNNs have recurrent hidden layers that serve as memory, which maintains temporal dependencies in the data.

RNNs are used in river water quality forecasting, including time-series forecasting [144]. RNNs predict future water quality indicators based on historical data. Meanwhile, they are used to predict algae bloom. The algae bloom can be used to reveal the trend of eutrophication in the river [145] by modeling the progression of nutrient levels and their impacts on algal growth over time. RNNs also integrate real-time sensor data to provide dynamic predictions of water quality parameters.

The most apparent advantage of RNNs is that they are temporal dependency models. The recurrent structure allows RNNs to account for the order of water quality events, making them ideal for time-series prediction tasks. RNNs can process multiple water quality parameter inputs simultaneously to predict various outputs.

3.2.3. Convolutional Neural Networks

CNNs excel at extracting spatial patterns and features from structured data, making them efficient for tasks like analyzing satellite imagery of rivers, detecting pollution sources, and studying spatial variations in water quality [146].

CNNs are mainly used for spatial pattern analysis. They analyze spatial variations in water quality parameters by processing gridded datasets [147]. CNNs have a strong capacity for image analysis [148,149]. In this case, they can analyze the satellite imagery [150]. The satellite images contain pollution sources, and algae blooms are used to analyze river water quality [146]. CNNs can combine spatial data from sensors to predict water quality at different locations in a river system.

For CNNs, they automatically learn spatial patterns and relationships, reducing the need for manual feature engineering. CNNs efficiently handle large datasets, including high-resolution imagery and multi-sensor spatial data. They are also robust to noise and variations in the data, making them well-suited for diverse environmental conditions.

3.2.4. Long Short-Term Memory Networks

By incorporating memory cells and gates, LSTMs can effectively capture both short-term and long-term dependencies. There is a special type of LSTM that shows an advanced ability for river water quality forecasting, called gated recurrent units (GRUs) [127,151–154]. GRUs are similar to LSTM networks but with a simplified architecture [155]. LSTMs and GRUs are particularly effective for river water quality prediction, where temporal relationships are essential, and computational efficiency is often required.

Both LSTMs and GRUs are special RNNs. The main difference with them is that the cells' structures are different. The information derived from the previous cells will be new for the following cells. The equations for LSTMs are shown below:

$$f_t = \sigma\left(W_f[h_{t-1}, x_t] + b_f\right) \tag{2}$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \tag{3}$$

$$\widetilde{C}_t = \tanh(W_C[h_{t-1}, x_t] + b_C) \tag{4}$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \widetilde{C}_t \tag{5}$$

$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o) \tag{6}$$

$$h_t = o_t \otimes \tanh(C_t) \tag{7}$$

where f_t is the forget gate output; i_t and o_t are the outputs for the input and output gate, respectively; t is the time step; C_t is the memory cell state; $\widetilde{C}_t$ is the candidate cell state; W_f, W_i and W_C represent the weight matrix for the forget, input and output state, respectively; h is the hidden state; x is the current input; b_f, b_i and b_C are the bias for different gates; $\sigma(\ )$ is the sigmoid activation function.

For GRUs, the equations are similar. The equations can be represented as

$$z_t = \sigma(W_z[h_{t-1}, x_t] + b_z) \tag{8}$$

$$r_t = \sigma(W_r[h_{t-1}, x_t] + b_r) \tag{9}$$

$$\widetilde{h}_t = \tanh(W_h.[r_t \otimes h_{t-1}, x_t] + b_h) \tag{10}$$

$$h_t = z_t \otimes h_{t-1} + (1 - z_t) \otimes \widetilde{h}_t\Big) \tag{11}$$

where z_t is the update gate output; r_t is the reset gate output. $\widetilde{h}_t$ is the candidate activation. The other letters have the same meanings as LSTM equations.

LSTMs and GRUs are robust to sequential data challenges [155]. They overcome the vanishing gradient problem, allowing for stable training and improved performance on long time-series data. They are suitable for real-time applications where continuous data streams from sensors need to be analyzed dynamically. As they have different structures, they are also advanced in different fields. Table 2 lists the different characteristics of LSTMs and GRUs.

Table 2. The comparison between LSTMs and GRUs.

Feature	LSTMs	GRUs
Parameter Complexity	More parameters	Fewer parameters
Computational Efficiency	Slower due to complexity	Faster due to simplicity
Performance	Better for long sequences	Comparable for shorter sequences
Flexibility	Offers more control over memory management	Simpler and easier to implement
Application	Capturing the effects of climate variability or upstream pollution events over extended timeframes	Modeling short-term impacts of rainfall events on downstream water quality

Models excel in sequential data analysis and are useful for water quality prediction. LSTMs are more powerful for analyzing long-term information but are computationally heavier. GRUs are simpler, faster, and more effective for tasks with moderate temporal dependencies.

3.2.5. Transformer Models

Transformer models have been applied for forecasting water quality in the recent years [156,157]. The self-attention mechanism is a commonly used transformer model [158,159]. It enables the model to concentrate on the important sections of the input sequence when generating predictions. Multi-head attention models are also a kind of transformer model [160]. They enhance the model's ability to recognize various patterns by utilizing multiple attention mechanisms simultaneously.

Transformer models allow themselves to capture relationships over long sequences without the vanishing gradient problem [161]. They process all input elements simultaneously, significantly speeding up training and inference. Transformers can also integrate multivariate data, including temporal, spatial, and contextual information, making them versatile for water quality prediction tasks. The attention weights provide insights into the input sequence that is most relevant to the prediction. This improves the explainability.

Time-series forecasting and spatiotemporal analysis are the applications of transformers in river water quality prediction [162]. They forecast water quality indicators based on historical data with time and space information. They can also identify and attribute water quality changes to specific pollution sources using multivariate data.

Deep learning models have significantly advanced the river water quality prediction field by offering superior capabilities in handling large, complex, and unstructured datasets. Each model contributes unique strengths to water quality prediction, providing researchers with flexible tools to address various aspects of river system dynamics. The integration of temporal, spatial, and multivariate data into deep learning models has made them indispensable for modern environmental management.

3.3. Hybrid Machine Learning Models

Hybrid machine learning models combine the strengths of different modeling techniques, including traditional machine learning, deep learning, and statistical methods, to enhance predictive accuracy, robustness, and adaptability. These models are particularly valuable in river water quality prediction, where the data are often complex, multivariate, and span both temporal and spatial dimensions. By leveraging multiple approaches, hybrid models can address the limitations of individual models and improve performance across diverse tasks.

The hybrid machine learning models combine different models. One example is the integration of statistical and machine learning methods. Statistical models capture linear trends, while machine learning models address non-linear relationships [30,163–166]. Combining different machine learning models is commonly used to develop hybrid models. As an example, hybrid models often combine algorithms such as random forests, SVMs, and gradient boosting with deep learning techniques like LSTMs or CNNs [167,168]. Spatiotemporal models can also be combined [168–172]. Ensemble approaches, such as stacking, boosting, or bagging, combine multiple models to improve generalization and reduce errors.

The hybrid machine learning models are complex but more accurate than simple models. Combining models helps to capture both linear and non-linear relationships, leading to more precise predictions. Meanwhile, hybrid models are able to adapt to a variety of data types and tasks. For example, CNNs are not good at dealing with temporal datasets. If we use CNN-LSTM, this hybrid model can deal with spatial and temporal datasets [173]. Besides this, hybrid models do not easily overfit when they are dealing with noisy datasets.

3.4. Overview of the Useful Machine Learning Method for River Water Quality Forecasting

River water quality forecasting has evolved significantly over the years, moving from traditional monitoring techniques to advanced data-driven approaches that leverage machine learning and artificial intelligence. Early efforts primarily focused on manual sampling and empirical statistical methods to predict water quality parameters like DO, BOD, and nutrient levels [64]. These methods, while foundational, were limited in their ability to handle complex datasets or predict water quality in dynamic environments.

The advent of machine learning marked a turning point in water quality forecasting. Techniques such as ANNs began to be applied, enabling the modeling of non-linear relationships and improving prediction accuracy. ANNs demonstrated their ability to forecast key parameters such as turbidity, pH, and TSS, but their reliance on high-quality datasets and computational resources presented challenges.

In recent years, hybrid models combining machine learning algorithms with data preprocessing techniques have gained prominence. For instance, hybrid decision tree-based models have been shown to enhance short-term water quality predictions by reducing noise and capturing critical features. These models, integrating tools like wavelet transforms and ensemble learning, have significantly improved the robustness of predictions, especially in rivers affected by urbanization and industrial activities.

Deep learning further revolutionized the field, with models such as LSTMs and CNNs widely adopted for spatio-temporal water quality forecasting. These models capture temporal dependencies and spatial heterogeneity, as demonstrated in studies focusing on complex river systems like the Burnett River in Australia and the Kelantan River Basin in Malaysia [174].

More recently, transformer models have emerged as a cutting-edge approach in water quality forecasting, offering advanced capabilities in handling long-term dependencies and large datasets. For example, the temporal fusion transformer has been applied to model and forecast water quality indexes in multi-site river networks, showcasing its potential for integrating diverse data sources [78].

The evolution of river water quality forecasting reflects a shift from manual and statistical methods to sophisticated machine learning and deep learning models driven by the availability of large datasets. Although there are advancements, challenges such as data quality, interpretability, and model generalization remain areas for future development.

Table 3 lists recent years' research on river water quality forecasting. According to Table 3, hybrid models are widely adopted by researchers, with LSTM-based models being the most commonly used. In recent years, real-time monitoring has become a prevalent method for data collection, resulting in datasets that are both temporal and spatial in nature. LSTM-based models, including those using GRU architectures, demonstrate strong capabilities in analyzing and forecasting such real-time datasets. Those models fulfil the modern water quality prediction tasks.

The most commonly used and predicted parameters are oxygen indicators, such as DO, BOD and COD. These parameters are frequently investigated due to their importance in maintaining aquatic life, with DO being particularly vital for the survival of aquatic organisms [48]. The table also shows that the nutrient level is the most common water quality parameter that researchers have assessed recently (especially in 2025). As for the previous research, most water bodies face eutrophication. This is mainly caused by substantial nutrients in the water, including ammonia, TP and TN.

Another key finding is that total coliform levels are influenced by multiple external factors, making precise prediction challenging. To address this complexity, advanced modeling techniques are required. Hybrid models integrate multiple predictive approaches and are the most effective choice for capturing these influencing factors and improving prediction accuracy.

Table 3. The summary of machine learning models used for water quality forecasting.

	A *	**B** *								**C** *			
	Temperature	**Discharge**	**Turbidity**	**DO**	**BOD**	**COD**	**TSS**	**TDS**	**Nutrient Levels**	**pH**	**Heavy metal**	**Algae**	**Total Coliform**
Traditional machine learning													
SVR (2015) [175]		✓											
ARIMA (2020, 2022) [176,177]				✓						✓			
ETR (2021) [165]				✓	✓	✓		✓					
Random Forest (2022) [178]			✓						✓				
XGBoost (2022) [128,179]			✓	✓	✓	✓							
NuSVC (2022) [180]										✓			
Regression (2025) [181]				✓	✓								
GWR (2025) [182]									✓				
Deep learning													
LMA (2009) [134]				✓	✓								
MLP (2019) [183]	✓		✓						✓	✓			✓
DNN (2020, 2022, 2023) [180,184,185]			✓	✓						✓	✓		
FNN (2021) [186]	✓												
LSTM (2022) [187]				✓	✓	✓	✓		✓	✓			
STNX (2024) [188]			✓	✓						✓			
LWQformer (2025) [189]				✓		✓				✓			
Hybrid machine learning													
CEEMDAN-XGBoost (2020) [190]			✓							✓			
CEEMDAN-RF (2020) [190]	✓			✓									
SWAT-ANN (2020) [191]								✓	✓			✓	
1-DRCNN-BiGRU (2021) [192]					✓				✓				
PCA-QRF (2021) [193]				✓									

Table 3. *Cont.*

	A *			B *							C *		
	Temperature	Discharge	Turbidity	DO	BOD	COD	TSS	TDS	Nutrient Levels	pH	Heavy metal	Algae	Total Coliform
SWAT-ANN (2020) [191]								✓	✓			✓	
1-DRCNN-BiGRU (2021) [192]						✓			✓				
PCA-QRF (2021) [193]					✓								
TrAdaBoost-LSTM (2021) [194]				✓									
SWT-ISSA-LSTM (2021) [195]				✓									
VMD-SSA-LSSVM (2021) [167]				✓					✓				
PSO-GEP (2021) [196]				✓				✓					
CNN-LSTM (2021) [197–199]				✓					✓	✓			
ANN-WT-LSTM (2022) [200]				✓		✓			✓				
SOD-VGG-LSTM (2022) [201]				✓		✓			✓				
DLBL-WQA (2022) [202]					✓	✓							
VMD-IGOA-LSTM (2023) [203]				✓		✓			✓				
SG-STL-TCN (2023) [204]				✓									
GJO+CNN-LSTM-TCN (2023) [205]									✓				
GNN-SAGE (2023) [206]		✓											
ANN-GBM (2023) [207]												✓	
ReliefF-GSA+ LightGBM (2024) [208]			✓						✓				
SVM-RBF (2024) [209]												✓	
CGAN-CA-CNN (2025) [210]									✓				
EMD-RBMO-SVMD-CNN-BiGRU (2025) [211]									✓				
LSTM-MOOTLBO (2025) [212]				✓									

* A: Physical indicators. B: Chemical indicators. C: Biological indicators.

Meanwhile, this provides more water quality indicators, which might help with accurate water quality forecasting. In this case, many indicators are used to find the source of pollution, like river basin-specific pollutants (RBSPs) [202,213–215], aromatic compounds [216] and other organic or inorganic substances of origin. Compared with other water quality indicators, they show specific characteristics and can be related to local point source pollution. Many data collection methods, such as satellite images, can also be related to diffuse pollution.

Table 4 compiles representative case studies, listing not only the predictive models and their associated performance metrics (e.g., RMSE, MAE, R^2), but also the corresponding climate zones and major anthropogenic stressors affecting river water quality. The selection of models for inclusion is guided by the need to capture a diverse range of climatic conditions and temperate continental zones.

Table 4. The climate zone and model performances of different models.

Reference	Model	Climate Zone	Impacted by	Best Predicted Parameters
Callegari et al. (2015) [175]	SVR	Cold alpine region	Seasonal river discharge; snow cover area	RMSE = 24%; R^2 = 0.81 (discharge)
Alnahit et al. (2022) [178]	Random Forest	Humid subtropical	Agriculture and urbanization	RMSE = 0.061; MAE = 0.022; NSE = 0.56 (nutrient levels)
Khoi et al. (2022) [179]	XGBoost	Tropical monsoon	Agriculture, urbanization and industry	RMSE = 0.107; R^2 = 0.989 (WQI)
Peterson et al. (2020) [184]	DNN	Temperate continental	Agricultural runoff	R^2 = 0.883 (Chl-a), 0.894 (DO), 0.839 (turbidity) RMSE = 7.561 (Chl-a), 1.806 (DO), 5.190 (turbidity) MAPE = 26.71 (Chl-a), 9.08 (DO), 9.87 (turbidity)
Feigl et al. (2021) [186]	FNN	Temperate	Climate-driven river temperature	RMSE = 0.422; MSE = 0.329 (temperature)
Yan et al. (2021) [192]	1-DRCNN-BiGRU	Temperate	Industrial and agricultural runoff	MAPE = 2.4866; R^2 = 0.9431 (nutrient levels)
Song et al. (2021) [167]	VMD-SSA-LSSVM	South subtropical monsoon	Precipitation and industry	MAE = 0.2178; RMSE = 0.2934; MAPE = 2.4637%; R^2 = 0.6351; NSE = 0.9541 (DO)
Wang et al. (2025) [209]	CGAN-CA-CNN	subtropical monsoon	Agriculture and industry	RMSE = 0.0019; MAE = 0.0009; R^2 = 0.9805 (nutrient levels)
Doroudi et al. (2025) [212]	LSTM-MOOTLBO	Subtropical dry summer	Urbanization	R = 0.967; RMSE = 0.303; NSE = 0.935 (DO)

Based on the analysis in Table 4, model effectiveness appears closely linked to environmental conditions. As rivers traverse diverse landscapes, they are subjected to varying climatic regimes and anthropogenic influences. Agricultural runoff emerges as a common and significant threat across many river systems. In cold regions, snowmelt processes introduce additional complexities by causing substantial variations in river discharge, which, in turn, affect water quality dynamics.

Typically, as rivers flow through different climatic zones, they are influenced by multiple interacting factors. However, most existing studies tend to focus on specific river segments or localized regions, which helps to reduce the confounding effects of mixed climatic and anthropogenic interactions.

Forecasting river water quality in a specific area can serve as a valuable reference for shaping future governmental strategies. Consequently, temporal and spatial–temporal water quality forecasting has recently gained increasing attention. Hybrid models are commonly employed to enhance predictive accuracy, leveraging the strengths of multiple approaches to enhance model performance.

4. Limitations of the Present River Water Quality Forecasting Methods

The river water quality models discussed in the previous section demonstrate strong performance. However, they still have certain limitations, some of which are influenced by environmental factors.

4.1. Disadvantages of Machine Learning Models for Forecasting

Challenges inherent to these models, such as suboptimal performance and overfitting, present significant issues for researchers in this section. Additionally, improving machine learning models' interpretability continues to be a critical area of concern.

4.1.1. Traditional Machine Learning Models

As mentioned above, the traditional machine learning methods are simple. This brings a lot of benefits. However, this also causes some limitations. Small and simple traditional machine learning methods struggle with high-dimensional, large-scale datasets or datasets with complex temporal and spatial structures. They are unable to handle complex features in the dataset. Based on the important water quality indicators section, there are many indicators that make the dataset non-linear [30,174,217]. Some simple traditional machine learning models, like linear regression, cannot capture non-linear interactions between water quality indicators.

4.1.2. Deep Learning Models

For deep learning models, ANNs restrict data requirements. They require large, high-quality datasets for training to avoid overfitting and ensure reliable predictions [136,218]. Training the ANN models requires significant time and equipment. ANNs models are black box models [130]. The models lack interpretability. This makes it hard for ANNs to understand how predictions are made. This can hinder trust in decisions based on ANN outputs. ANNs may memorize training data instead of generalizing, particularly when the dataset is small or noisy.

During backpropagation, gradients can shrink to near zero. Learning long-term dependencies is hard [141]. This limitation often necessitates using advanced RNN architectures like LSTM or GRU. There are many recurrent hidden layers in the RNNs. This increases computational complexity [219]. Training RNNs can be computationally demanding, particularly when dealing with long sequences and extensive datasets. RNNs require extensive and high-quality datasets for effective training.

CNNs have similar limitations to the previous models [219]. They also need large datasets for training to prevent overfitting and achieve generalization. CNNs are also limited temporal models. They are primarily designed for spatial data and lack the inherent ability to capture temporal relationships, requiring integration with time-series models like RNNs [133,173].

LSTMs and GRUs are useful machine-learning models for river water quality forecasting. There are still some disadvantages of the models. Both models require high-quality datasets for effective performance [219,220]. They require careful and time-consuming tuning for optimal performance [221]. If there are missing data and noisy datasets, they still cannot perform well.

Given the limitations of transformer models, they still need large datasets. As they analyze and forecast based on historical information, the limited and small dataset causes the model to not perform well, and the architecture is complex. This requires expertise in model design and hyperparameter tuning. The self-attention mechanism makes transformers memory-intensive for very long sequences [222].

Based on the previous analysis, the dataset is the biggest challenge for deep learning. Most of the models require high-quality datasets, and the neural network models are complex. This makes them expensive and raises the challenges of overfitting, as most of the models are black box models. It is challenging to understand how predictions are made.

4.1.3. Hybrid Machine Learning Models

Although hybrid machine learning models perform well, their complexity also increases. Designing and implementing hybrid models require expertise and the careful tuning of multiple components. This increases the cost as it increases computational resource requirements and training time. Integrating multiple models can make it difficult to explain predictions or identify the contributions of individual components.

4.2. Summary of the Machine Learning Models Used for River Water Quality

Based on the previous sections, we can use Table 5 to summarize the characteristics of river water quality forecasting. To put it simply, traditional machine learning models are ideal for simple physical indicators analysis, but lack the capacity to handle complex dynamic systems. Deep learning models excel in accuracy and spatial–temporal dynamics modeling, but are resource-intensive and challenging to interpret. They can learn and find the relationship between time and parameters. Hybrid models represent a key future direction, combining multiple approaches. Hybrid models represent improvements in the performance of simple deep learning models.

Table 5. The characteristics of machine learning (ML) models for river water quality forecasting.

	Traditional ML	**Deep Learning**	**Hybrid ML**
Advantages	Strong predictive ability for simple nonlinear relationships; highly adaptable	High accuracy, capable of learning spatial–temporal dynamics, suitable for large-scale, complex datasets	Highly comprehensive, significantly improves accuracy, ideal for addressing complex water quality issues
Disadvantages	Relies heavily on data quality; prone to overfitting; difficult to interpret	High computational cost; requires substantial resources; lacks transparency	Complex architecture, longer development cycles, requires advanced technical expertise
Common methods	Random Forest, SVM, XGBoost	NN-based model	Deep learning (like LSTM and GRU) models; integration of physical and statistical/ML models
Suitable forecast parameters	Heavy metal concentrations	Captures spatial–temporal dynamics, suitable for complex time-variant parameters like DO, BOD and COD	Comprehensive modeling, able to handle multi-dimensional parameters like discharge, and long-term trends

There are some challenges for the river water quality field. River water quality forecasting using current machine learning models faces several challenges that hinder its accuracy, reliability, and practical application. One significant issue is data availability and quality, as many regions lack comprehensive, high-resolution datasets due to limited monitoring infrastructure, leading to gaps and inconsistencies that affect model training and validation [223].

Additionally, handling complex spatiotemporal dependencies remains a challenge, as current models often struggle to simultaneously account for the interactions between

spatial pollution patterns and temporal variations in water quality indicators [224]. Model interpretability is another critical limitation. The black box function of machine learning makes it hard to know the rationale behind predictions and trust their outcomes [225]. Furthermore, computational costs associated with advanced models like deep learning can be prohibitive, especially for resource-constrained settings requiring real-time predictions [226]. Models also face difficulties in adapting to climate change impacts, introducing new uncertainties in water quality dynamics [164]. Finally, the generalizability of models across different river systems is limited, as each system's unique hydrological, geographical, and anthropogenic factors often necessitate extensive retraining and fine-tuning [174]. Addressing these challenges improves the effectiveness and adoption of machine learning models in river water quality forecasting.

4.3. Challenges in Data Collection for River Water Quality Forecasting

In addition to model-related challenges, data collection is another significant obstacle in river water quality forecasting. The factors contributing to river water quality deterioration are complex, with climate change and human activities being the primary drivers.

Extreme weather events, such as severe droughts and floods caused by climate change, significantly influence river discharge patterns, introducing considerable uncertainty into water quality forecasting [227,228]. Climate change also affects air temperature and wind patterns, further altering key water quality parameters [229,230]. Moreover, the inherent difficulty of accurately forecasting long-term weather conditions presents an additional layer of complexity, as these uncertainties directly impact the reliability of river water quality predictions.

Also, the river discharge brings another difficulty, called lag time. The lag time is the delay between pollutant input and observable changes in water quality parameters. Weather changes and agricultural runoff always cause a lag time. The lag time problem affects the accuracy of water quality forecasting [231].

Human activities like mining, industrial expansion, urbanization and agriculture introduce pollutants and alter river ecosystems, making water quality prediction increasingly complex. The interplay between diverse pollution sources and natural processes introduces nonlinearity and high variability in water quality parameters. This makes it challenging to develop models that can accurately account for both temporal and spatial variations [232]. Human activity also affects climate [233]. Global warming is a typical example. This further influences the prediction of river water quality.

While machine learning has greatly advanced river water quality forecasting, challenges remain in model reliability, data dependency, and computational costs. Traditional models struggle with complex, non-linear relationships, deep learning requires extensive data and resources, and hybrid models, though more accurate, are often complex and less interpretable. The following section explores these limitations, highlighting key data quality, model generalization, and scalability obstacles, and providing insights into future improvements.

5. Conclusions

In recent years, inland river water quality forecasting has significantly advanced, transitioning from traditional monitoring methods to hybrid remote-sensing, real-time monitoring and sophisticated machine learning-based approaches. While conventional monitoring methods provide critical insights, they are often labor-intensive, costly, and lack real-time forecasting with adequate spatial coverage to capture spatiotemporal variability. Machine learning techniques have emerged as powerful tools for water quality forecasting,

offering the ability to process large datasets, model non-linear relationships, and generate accurate predictions across spatial and temporal scales.

This review examined various machine learning approaches, including traditional models, deep learning techniques, and hybrid frameworks. The emergence of deep learning and hybrid models, particularly LSTMs, GRUs and LSTM, and GRU-based models, has improved the ability to capture spatiotemporal dependencies in river water quality data. These models have proven highly effective in handling large-scale datasets and improving prediction accuracy. However, they also pose challenges, including high computational costs, reliance on large training datasets, and limited interpretability. Current methods cannot provide long real-time forecasts to meet water management requirements for timely early warnings, and there is a lack of updated datasets relevant to improved forecast accuracy. All of this is based on the current literature and the gaps we have identified.

Hybrid deep learning models are widely used; however, their complexity often makes them difficult to interpret. Attention-based deep learning models provide a potential solution to this issue. For instance, STA-GRU models have been used for water-level forecasting, demonstrating strong predictive performance [221]. Additionally, graph neural networks (GNNs), which leverage graph topology [234], have been effectively used for river water quality forecasting by modeling the diffusion of pollutants.

Another direction is the development of hybrid modeling approaches, combining machine learning techniques with data preprocessing tools. Multi-source data integration is crucial, incorporating information from satellite imagery and hydrological models to build comprehensive datasets for improved spatial and temporal analysis [235]. The availability and completeness of datasets often limit advanced machine learning models. Integrating remote sensing imagery and monitoring data enhances dataset comprehensiveness, providing a more robust foundation for accurate water quality predictions. For human-affected, complex, non-linear datasets like mine water datasets, thermodynamic models have been employed to perform predictions in mining-impacted regions [236]. Combining these physics-based models with machine learning techniques could allow for more robust and adaptive water quality forecasting frameworks in complex geological and hydrogeological conditions.

Applying explainable AI is also vital for improving stakeholder trust and the interpretability of predictions, especially in decision-making scenarios that require transparency [237]. Techniques such as Shapley Additive Explanations (SHAP) [48,128,238,239] and Local Interpretable Model-Agnostic Explanations (LIME) [238,239] provide feature-level interpretability by quantifying the contribution of each input variable to the model's prediction.

Future models could focus on adapting to climate change impacts, such as increasing extreme weather events and shifting precipitation patterns, by incorporating dynamic inputs into predictive frameworks [240]. Moreover, real-time forecasting systems powered by edge computing and advanced neural networks, such as CNNs and LSTMs, could enhance responsiveness and accuracy in operational water quality monitoring [241].

Finally, one potential avenue for further improvement is to enhance existing machine learning methods while optimizing the data collection process. This could involve leveraging remote sensing technology for more efficient data acquisition and integrating advanced weather forecasting models to obtain highly accurate meteorological information. Beyond technical improvements, future research could also expand the range of water quality indicators and the scope of forecasting applications. In particular, these approaches could be extended to tidal rivers and estuarine environments. By addressing both methodological enhancements and broader application domains, this review is expected to serve as a reference for developing more resilient and adaptive forecasting systems under increasing

climatic and anthropogenic pressures. It contributes both to academic discourse and to the practical foundation for deploying intelligent water quality monitoring systems worldwide.

Author Contributions: Conceptualization, D.P.; methodology, D.P.; formal analysis, D.P.; investigation, D.P.; resources, D.P.; writing—original draft preparation, D.P.; writing—review and editing, Y.D., S.X.Y. and B.G.; supervision, S.X.Y. and B.G.; project administration, S.X.Y. and B.G.; funding acquisition, S.X.Y. and B.G. All authors have read and agreed to the published version of the manuscript.

Funding: This research was funded by the Natural Sciences and Engineering Research Council of 535 Canada (NSERC) Alliance Grant #401643.

Data Availability Statement: No new data were created or analyzed in this study. Data sharing is not applicable to this article.

Conflicts of Interest: The authors have no conflicts of interest.

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